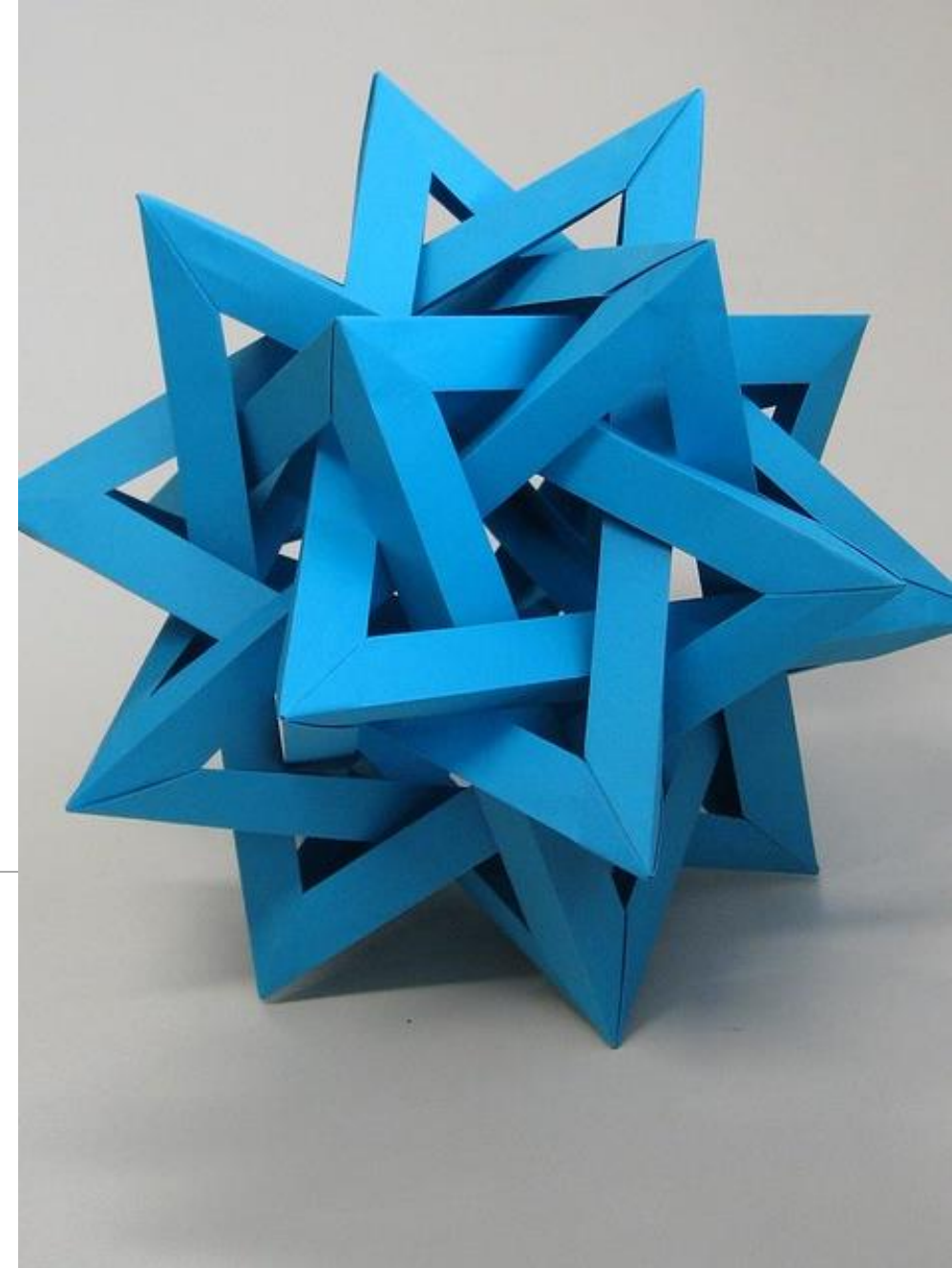


LABORATORY 11

INPUT FILES AVAILABLE IN THE
ZIP ARCHIVE [LAB11-INPUT-FILE.ZIP](#)



Recommended exercises

11.1.3

11.1.4

11.1.5

11.2.1

IMPORTANT: Familiarize with Exam Environment

- Start from **11.1.4** and solve it in the **Moodle platform**
- **Same environment you will find at the final exam.**
- See next slides for tutorial...

Access Simulation on Moodle

- From your course page, open the “Moodle” tab.
- Select the “Exam Simulation”
- Start your Attempt

Computer sciences

Course Grades Competencies

▼ General

Collapse all



[Prof. Pagliari] LAB3 - Peer Review #1



Not available unless: You enter the correct password



[Prof. Pagliari] LAB7 - Peer Review #2



Not available unless: You enter the correct password



Exam Simulation (Visual Studio Code and Crownlabs Test) - 2025

Mark as done



Structure of the Moodle Quiz

- This will open a **very similar environment** to what you'll have at the final exam.
- The first 2 pages contain “questions” that allow you to **withdraw** and to decide whether you want to use your **midterm** score.
 - Of course, in this simulation they are just there to show you how they'll look like. You can leave them blank...

Question 1
Not graded
Flag question

The Exam platform does not explicitly provide the option to withdraw from the exam. This question serves precisely for that purpose. Therefore, to indicate your intention to withdraw from the exam, simply write **"withdrawn"** as the answer to this question. If the answer is left blank, the exam will be graded.

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Question 2
Not graded
Flag question

If you want us to use your midterm score and ignore your answers to the three open-ended questions, please type **"use midterm score"** below. If you don't write anything, the entire exam will be corrected and the midterm score will **not be used!**

Later requests to reconsider this will not be accepted.

Structure of the Moodle Quiz

- Next, you have **3 questions** worth **2 points** each. **These are the ones you can skip** at the exam, if you decide to use your midterm score.
- In this simulation, we added three placeholder questions, similar to those you might get. You can try to answer in the text boxes below, or leave them blank...

Question 3
Marked out of 2,00
Flag question

Convert the decimal number 45_{10} to binary and then to hexadecimal. Write down all the steps to obtain the solution.

Question 4
Marked out of 2,00
Flag question

Describe the differences between volatile and non-volatile memory. Provide an example for each type.

Question 5
Marked out of 2,00
Flag question

What are the differences between Lists and Tuples in Python?

Structure of the Moodle Quiz

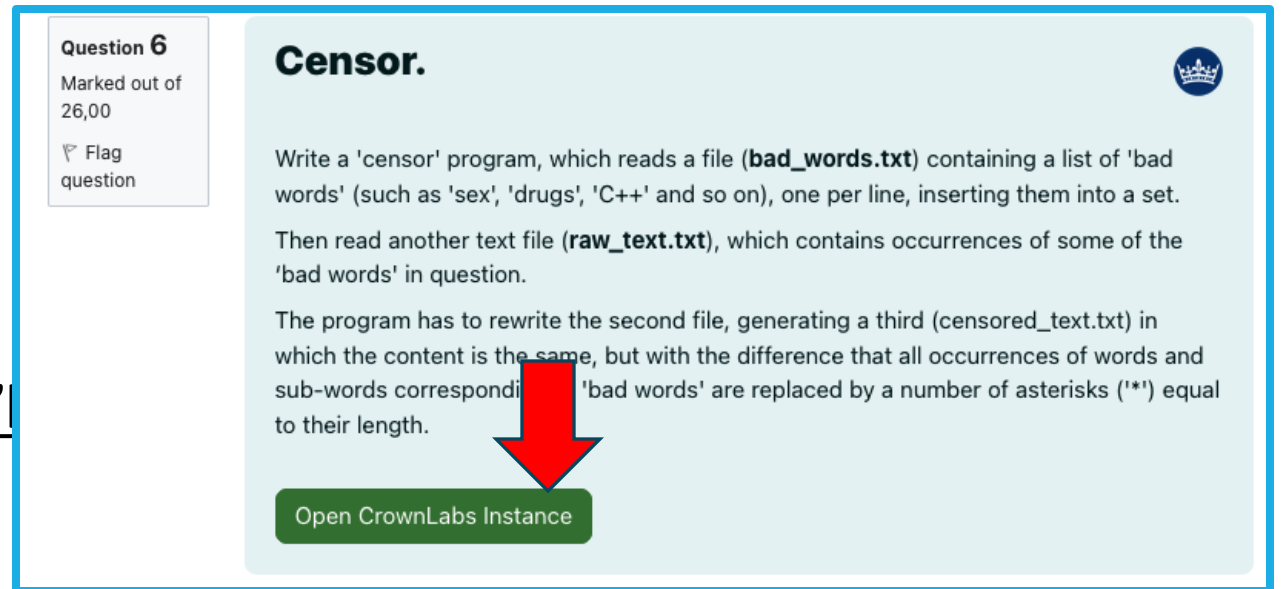
- The following page gives you access to various **documentation sources**:
 - The official Python documentation website.
 - A cheat sheet with commonly used Python functions and methods (also available in accessibility mode).
- **During the exam, Internet will be disabled except for the Python docs. So practice programming using only these two sources!!**

Information
🚩 Flag
question

Online Python Documentation (python.org)
CheatSheet PDF
CheatSheet (Accessible Version): ...will be added for the final exam

Structure of the Moodle Quiz

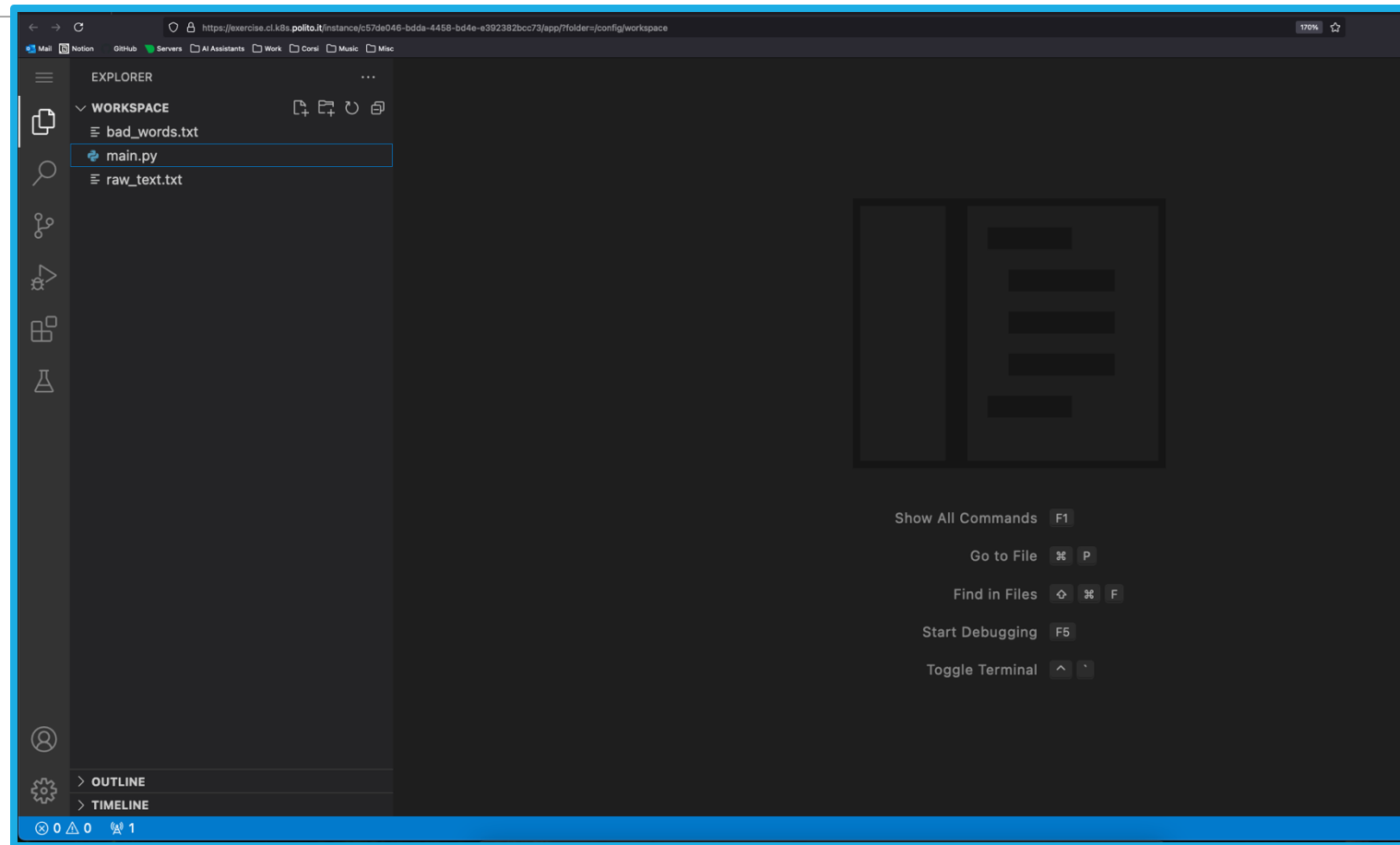
- The last page contains the text of the **programming exercise**. Currently, it is set to Ex. 11.1.4 (but of course the real exercise at the exam will be more complex!)
- Click on the button “Open CrownLabs Instance”.
- An **instance of Visual Studio Code will open inside your browser**
- This is the coding environment you’ll have to use at the exam.



The screenshot shows a Moodle quiz interface. On the left, a box for 'Question 6' indicates it is 'Marked out of 26,00' and has a 'Flag question' option. The main content area is titled 'Censor.' and contains the following text: 'Write a 'censor' program, which reads a file (**bad_words.txt**) containing a list of 'bad words' (such as 'sex', 'drugs', 'C++' and so on), one per line, inserting them into a set. Then read another text file (**raw_text.txt**), which contains occurrences of some of the 'bad words' in question. The program has to rewrite the second file, generating a third (censored_text.txt) in which the content is the same, but with the difference that all occurrences of words and sub-words corresponding to 'bad words' are replaced by a number of asterisks ('*') equal to their length.' At the bottom of the question area is a green button labeled 'Open CrownLabs Instance'. A large red arrow points from the text 'all occurrences of words and sub-words corresponding to 'bad words' are replaced by a number of asterisks' down to the button.

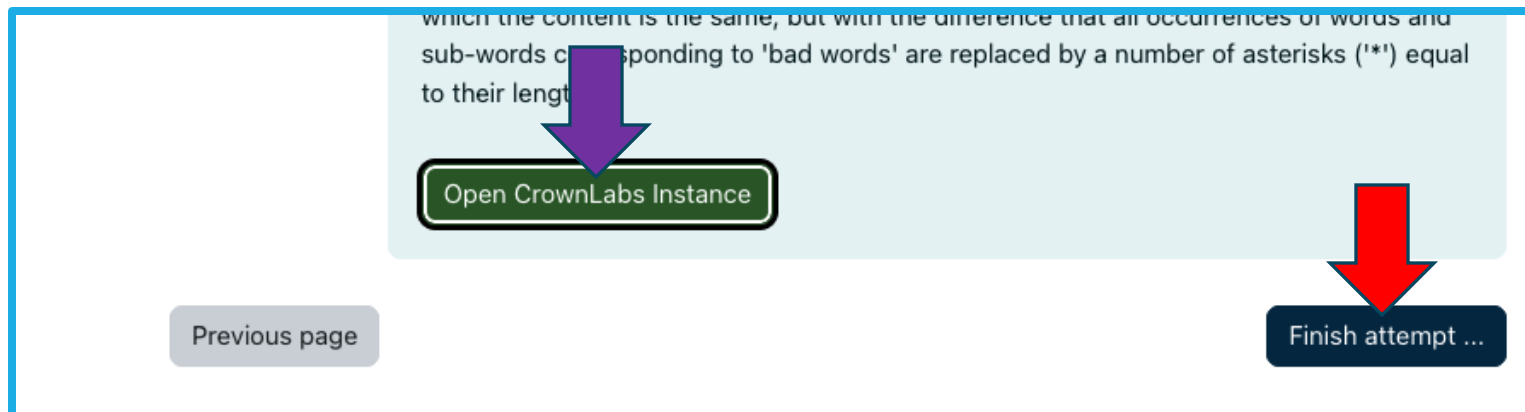
Structure of the Moodle Quiz

- You will find an empty Python file as well as your input files already there.
- **Do not create additional Python files, write your code only in the existing Python file!**
- Otherwise, **your work will not be saved** and we won't be able to evaluate it.



Structure of the Moodle Quiz

- You can run and/or debug your code as many times as you want.
- When you're done, simply close the VSCode window. If you need to, you can **re-open it** by clicking on the **"Open CrownLabs Instance"** button again.
- When you want to **conclude** your attempt, in the Moodle window, click on **"Finish Attempt"**.
- **NOTE: In this simulation, code you write in this VSCode instance will not be sent to us for evaluation. So, if you need it, remember to copy paste it somewhere else before you exit....**



Structure of the Moodle Quiz

- In the next window, you can check the status of your answers, go back to the Attempt, or submit all and Finish.

Exam Simulation (Visual Studio Code and Crownlabs Test) - 2025
Summary of attempt

Question	Status
1	Not yet answered
2	Not yet answered
3	Not yet answered
4	Not yet answered
5	Not yet answered
6	Answer saved

[Return to attempt](#)
[Submit all and finish](#)

Structure of the Moodle Quiz

- Once you submit, you will be automatically taken on a page where you can review your attempt.
- Click on "Finish review" to exit.
- **NOTE: In this simulation, code you write in the VSCode instance on Moodle will not be sent to us for evaluation. So remember to copy paste it somewhere else before you exit....**

DISCLAIMER

- While the final exam's environment will be very similar to this simulation, there might be some slight differences....
 - Example: the order of the different “pages” might change....
- **Always keep your brain “on”!!!**
- **Example: don't write the answer to one of the theory questions in the withdrawal box, etc...**

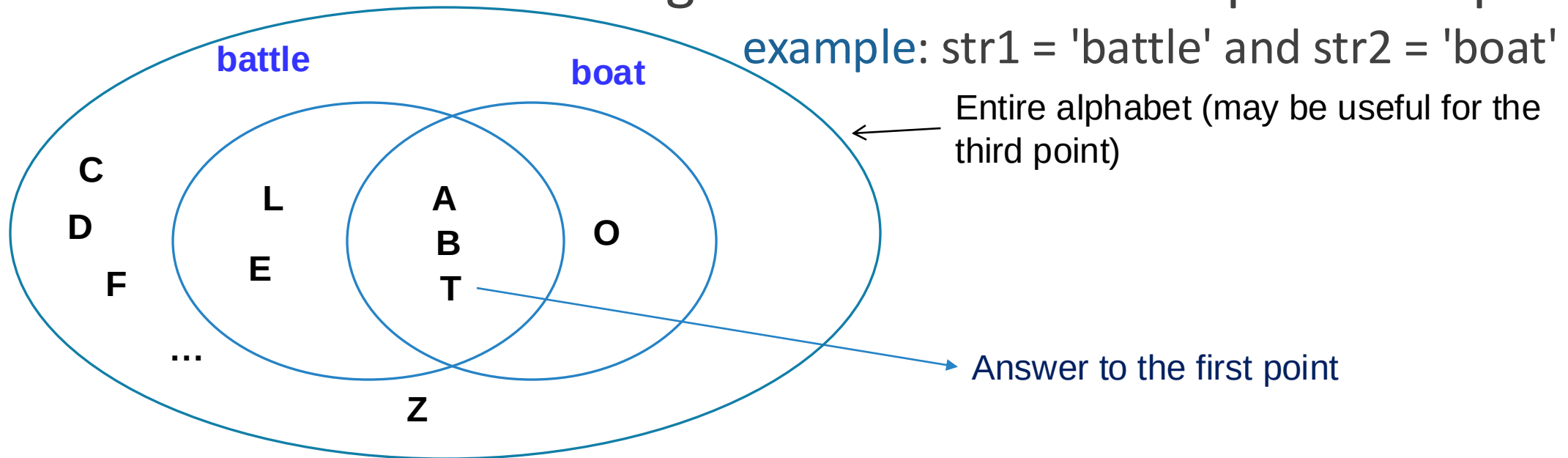
Exercise 11.1.3

Read request from lab text...

Exercise 11.1.3

Handle uppercase and lowercase letters (e.g. 'a' and 'A' must be considered the same character)

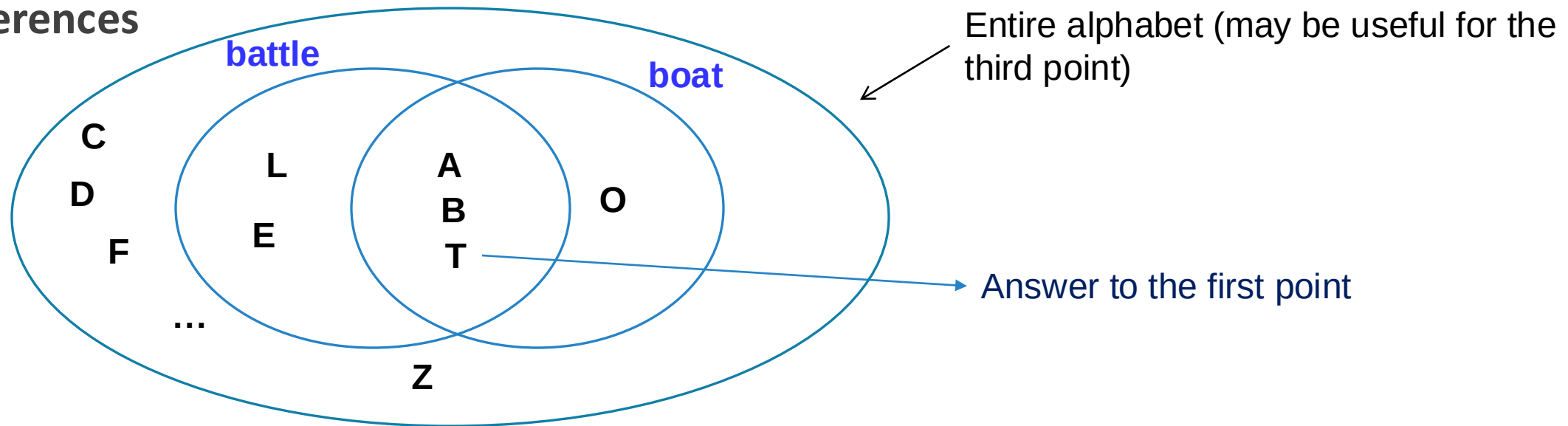
Use **the set methods** to generate the three required outputs



Exercise 11.1.3

Draw the sets on paper to verify what you're coding:

- Point 1: Simple
- Point 2 : There is a specific set operation to do this
- Point 3 : **We will have to define the “universe” of letters** in order to use set differences



Exercise 11.1.4

Read request from lab text...

Exercise 11.1.4

Which data structures?

- For the words to censor, we could use a list, but we know that a set is going to be more efficient to search through
- **Is it necessary store the rest of the data (the entire text to process?)**

Exercise 11.1.4

Program structure :

- **Read the file line by line**
- If a line contains a word to censor, replace it in the output file with as many asterisks
- **WARNING: Even substrings should be intercepted!**
- **For this you can use the `find()` method of strings**
- The rest of the text is just copied as is in the output.

PAY ATTENTION TO THE FORMAT!!!

- In the output file, we would like preserve the line structure of the input file...!

Exercise 11.1.5

Read request from lab text...

Exercise 11.1.5

The function receives the two dictionaries to be added and will return one that is the sum of the two.

Do not modify the input dictionaries

The algorithm “in English” is as follows:

- If a key is present **in both a and b** , add the key to the sum dictionary, and associate it with the value given by the sum of the values stored in **a** and **b**
- If a key is present **in only one of the two** dictionaries, add the key to the sum dictionary and associate it with the value stored in the dictionary that contains it.

Exercise 11.2.1

Read request from lab text...

Exercise 11.2.1

There are no particular difficulties: we saw a similar exercise in class...

It's a matter of separating each line by `splitting it` , and adding the country to the dictionary.

Let's try to handle the various errors that may be present in the data or in the user's requests

- **File with the correct number of fields**
- **Non-existent nations**

Exercise 11.2.1

Beware of possible format errors in the file:

- Does the file exist? (Try ... except)
- Are there two fields in each row or are some elements missing? (len ()!=...)

Unless otherwise specified, possible errors should be handled by the code!

NOTE: In this case no other statistics are requested, but in general it is advisable to convert numeric fields from strings to float, int ... (be careful of possible conversion errors!)

What happens if the user requests a country name that is not in the file?
Ask for another input in this case.